

Sub-task 1 - Create a Lambda with Asynchronous Invocation (Polling Invocation)

1. Create a Lambda function named <ProjectName>-UploadsNotificationFunction:
 - a. use any preferred programming language
 - b. grant the function the basic Lambda permissions as well as permissions for sending messages to <ProjectName>-UploadsNotificationTopic from the Module 9: SNS/SQS
2. Remove the code from the web-application that transfers SQS messages to SNS topic.
3. Add a Lambda trigger:
 - a. make sure that execution role of the Lambda contains permissions for the <ProjectName>-UploadsNotificationQueue from Module 9: SNS/SQS
 - b. choose <ProjectName>-UploadsNotificationQueue as SQS source
 - c. update the lambda code and implement sending messages to the <ProjectName>-UploadsNotificationTopic topic (this logic was removed from your web-application)
4. Make 2 or more image uploads using your web application. Ensure you got exactly emails with all the publications you've just made.

The screenshot displays the AWS Lambda console for the function 'cloudx2025-UploadsNotificationFunction'. The breadcrumb navigation shows 'Lambda > Functions > cloudx2025-UploadsNotificationFunction'. The function name is 'cloudx2025-UploadsNotificationFunction'. A notification bar states: 'This function belongs to an application. [Click here](#) to manage it.' The 'Function overview' section is active, showing a 'Diagram' tab with a visual representation of the function being triggered by 'SQS'. The 'Template' tab is also visible. On the right, the 'Description' section shows the function's ARN: 'arn:aws:lambda:ap-south-1:518382206174:function:cloudx2025-UploadsNotificationFunction'. The 'Application' is listed as 'cloudx2025-appstack'. The 'Function URL' is also present. The bottom of the screen shows the Windows taskbar with the date '24-09-2023' and time '13:02'.

aws Search [Alt+S] Asia Pacific (Mumbai) cloudx2025 (5183-9220-6174) cloudx_2025_admin

Amazon SNS > Topics > cloudx2025-UploadsNotificationTopic

cloudx2025-UploadsNotificationTopic

Edit Delete Publish message

Details

Name cloudx2025-UploadsNotificationTopic	Display name -
ARN arn:aws:sns:ap-south-1:518382206174:cloudx2025-UploadsNotificationTopic	Topic owner 518382206174
Type Standard	

< Subscriptions Access policy Data protection policy Delivery policy (HTTP/S) Delivery status logging Encryption >

Subscriptions (2) Edit Delete Request confirmation Confirm subscription Create subscription

Search

ID	Endpoint	Status	Protocol
0b71662c-5985-4c10-9c4d-811...	hmanak@gmail.com	Confirmed	EMAIL-JOIN

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Home Workspaces API Network Search Postman Ctrl + K Invite Upgrade

cloudx2025-app /images/upload

POST {{cloudx-app-url}}/images/upload Send

Params Authorization Headers (8) Body Scripts Settings Cookies

none form-data x-www-form-urlencoded raw binary GraphQL

Key	Value	Description
file	Mother_Seraphina_sketch.jpg	

Body Cookies Headers (4) Test Results Save Response

JSON Preview Visualize

```
1 {
2   "name": "Mother_Seraphina_sketch.jpg",
3   "s3Key": "1b9b06b9-01a1-4b63-b82a-b219217969ff-Mother_Seraphina_sketch.jpg",
4   "contentType": "image/jpeg",
5   "size": 239124,
6   "fileExtension": ".jpg",
7   "uploadedAt": "2025-09-24T07:28:15.697662416",
8   "lastUpdatedAt": "2025-09-24T07:28:15.697662416"
9 }
```

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AWS Notification Message



AWS Notifications <no-reply@sns.amazonaws.com>
to me

12:58 PM (0 minutes ago) ☆ 😊 ↶ ⋮

```
{
  "Type": "Notification",
  "MessageId": "3aafa1bc-4362-58ff-8d34-c0f1dcb2b943",
  "TopicArn": "arn:aws:sns:ap-south-1:518382206174:cloudx2025-UploadsNotificationTopic",
  "Message": "Image uploaded: Mother_Seraphina_sketch.jpg (239124 bytes, jpg). Download: /images/download/Mother_Seraphina_sketch.jpg",
  "Timestamp": "2025-09-24T07:28:16.675Z",
  "SignatureVersion": "1",
  "Signature": "qsxbVyzhVcq0FwNcMS2zig7TKhY38h8ZPhAsRWpsXGt8EoDiYZKj09OLDIZsOG9B60Iak4HbuqELs8DyhCfoSHvh1Ud4Q00QLFgYVXIKzq50glQa8o03ilyingXioEHnkaht3nlwF6wXkKNsHaYzkk203oCTYppDfjN7JE6LAbcz6hvcNyzMudYteTvsZE88WNmXzrD/eqPLrVnOs1B8kQNI4E9H8Fe3AqyDrdcXGIY3mdp8Xwi+7mlP3NqsUkixsblZ8AXMMPQVWvgcbB/bc+p1r3ZjzEOVW0CinN78ZAtxX2pMq0YISfxQZxRaVDdnPp1oy9tH3epElkCBQ==",
  "SigningCertURL": "https://sns.ap-south-1.amazonaws.com/SimpleNotificationService-6209c161c6221df56ec1eb5c821d112.pem",
  "UnsubscribeURL": "https://sns.ap-south-1.amazonaws.com/?Action=Unsubscribe&SubscriptionArn=arn:aws:sns:ap-south-1:518382206174:cloudx2025-UploadsNotificationTopic:0b71662c-5985-4c10-9c4d-811641778084",
  "MessageAttributes": {
    "extension": {
      "Type": "String",
      "Value": ".jpg"
    }
  }
}
```

↶ Reply

↷ Forward



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cloudx2025-app / /images/upload

POST

{{cloudx-app-uri}} /images/upload

Send

Params

Authorization

Headers (8)

Body

Scripts

Settings

none

form-data

x-www-form-urlencoded

raw

binary

GraphQL

	Key		Value	Description	Bulk Edit
☒	file	File	image1.jpg		
	Key	Text	Value	Description	

Body

Cookies

Headers (4)

Test Results

200 OK · 133 ms · 371 B

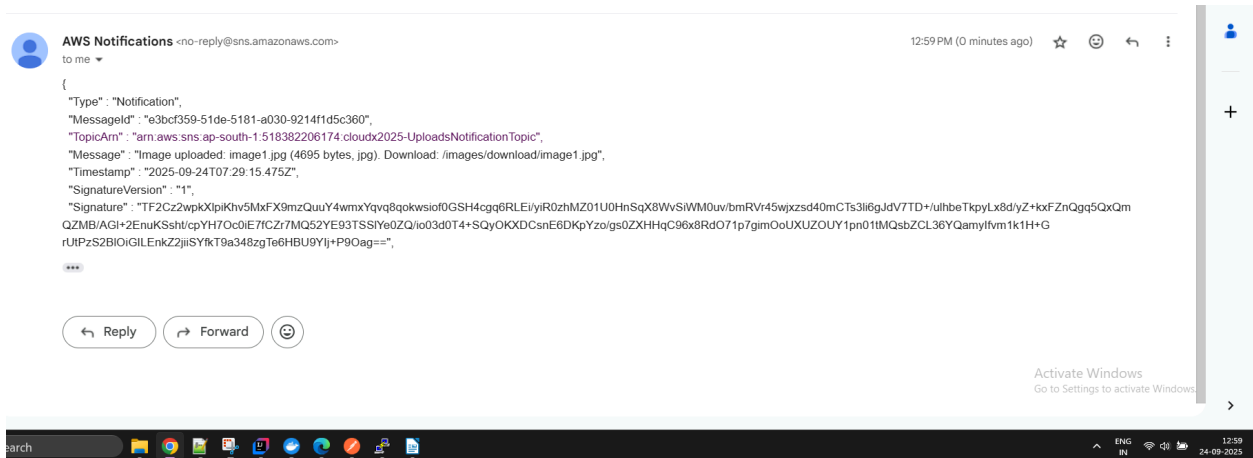
JSON

Preview

Visualize

```
1 {
2   "name": "image1.jpg",
3   "s3Key": "5eed47ef-1284-4e3e-baa4-8e26e57950e4-image1.jpg",
4   "contentType": "image/jpeg",
5   "size": 4695,
6   "fileExtension": ".jpg",
7   "uploadedAt": "2025-09-24T07:29:14.72807329",
8   "lastUpdatedAt": "2025-09-24T07:29:14.72807329"
9 }
```

RunnerStart ProxyCookiesVaultTrash



Sub-task 2 - Create a Lambda with Synchronous Invocation

- Create a Lambda function named <ProjectName>-DataConsistencyFunction:
 - use any preferred programming language
 - grant the function the basic Lambda permissions as well as permissions to have access to the DB from Module 8: RDS and to the S3 bucket from Module 4: S3
- Important: To ensure secure and private communication:
 - Create a VPC Endpoint for S3 to allow Lambda to access the S3 bucket without traversing the public internet.
 - Ensure the Lambda function resides in the same VPC as the RDS instance so it can securely access the DB.
- Implement some basic validation that verifies that DB metadata corresponds to the images in the S3 bucket (you can add some simple validation like finding image by name). The lambda should return a result that shows if data is consistent or not.
- Important Please keep in mind best practices of writing lambda code. Initializing connection to the DB should be outside the function.
- Create an endpoint in your web application which would do the same – simply trigger the <ProjectName>-DataConsistencyFunction Lambda.
- Follow this tutorial to make your Lambda run periodically (let's say, every 5 minutes).
- Make your Lambda logs scheduled/API Gateway/web application calls in such a way that it's possible to distinguish them in the CloudWatch logs. Hint – the tutorial from step 6 makes the Lambda receive messages which have a detail-type field, so it's possible to fill that field differently for the calls made from the web application and API Gateway endpoint.
- Make sure you can see all 3 (scheduled/API Gateway/web application) kinds of Lambda calls in the CloudWatch logs.

API Gateway - S x CloudFormation x CloudFormation x cloudx2025-Dat x cloudx2025-Dat x CloudWatch | ap x data_consistenc x Copy and Paste x +

ap-south-1.console.aws.amazon.com/lambda/home?region=ap-south-1#/functions/cloudx2025-DataConsistencyFunction?tab=code

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Lambda > Functions > cloudx2025-DataConsistencyFunction

Successfully updated the function cloudx2025-DataConsistencyFunction.

Function overview Info

Diagram Template

cloudx2025-DataConsistencyFunction

Layers (1)

EventBridge (CloudWatch Events)

API Gateway

+ Add trigger

Related functions: Select a function

+ Add destination

Description

Last modified 20 minutes ago

Function ARN: arn:aws:lambda:ap-south-1:518382206174:function:cloudx2025-DataConsistencyFunction

Application: cloudx2025-appstack

Function URL Info

Code Test Monitor Configuration Aliases Versions

Package size 1.2 kB

SHA256 hash: JMuHfL7jTxCsa90bWRJhRjuRQwbTnXD5bjvg2ufmq+8=

Last modified 20 minutes ago

Encryption with AWS KMS customer managed KMS key Info

Runtime settings Info

Runtime: Python 3.11

Handler: data_consistency.lambda_handler

Architecture: x86_64

Edit Edit runtime management configuration

Runtime management configuration

Layers Info

Edit Add a layer

Merge order	Name	Layer version	Compatible runtimes	Compatible architectures	Version ARN
1	cloudx2025-DataConsistencyLayer	6	python3.11	x86_64	arn:aws:lambda:ap-south-1:518382206174:layer:cloudx2025-DataConsistencyLayer:6

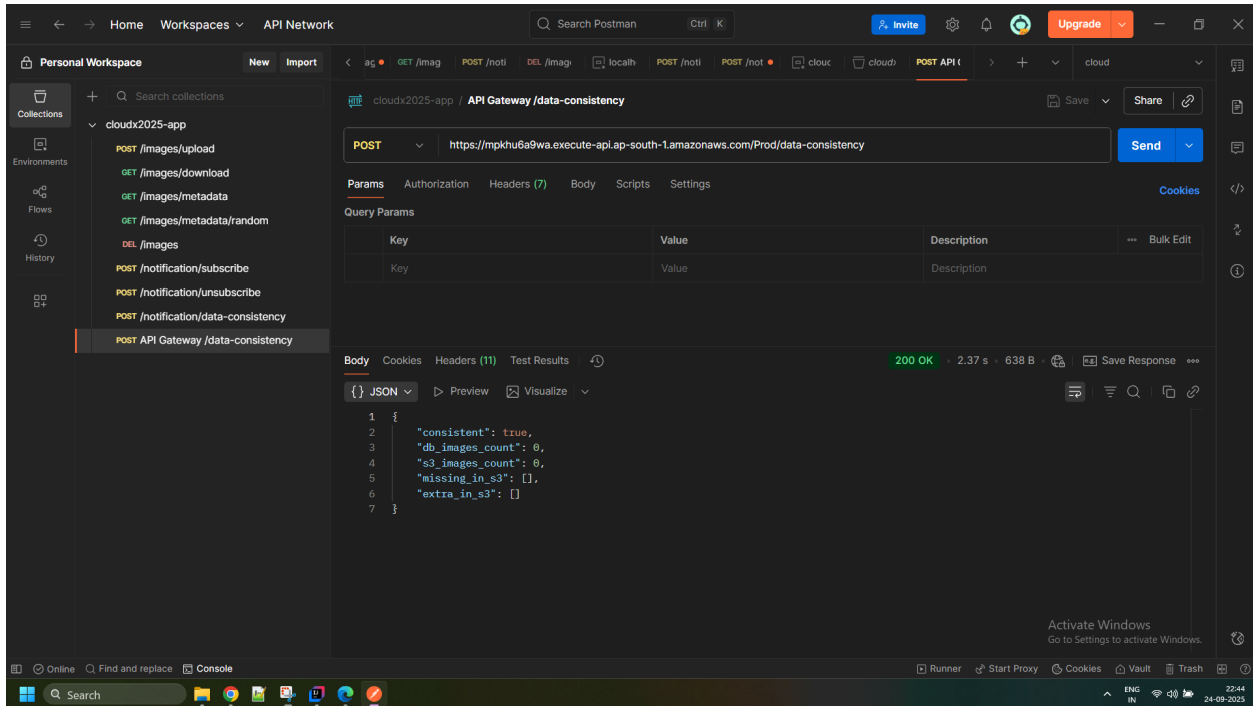
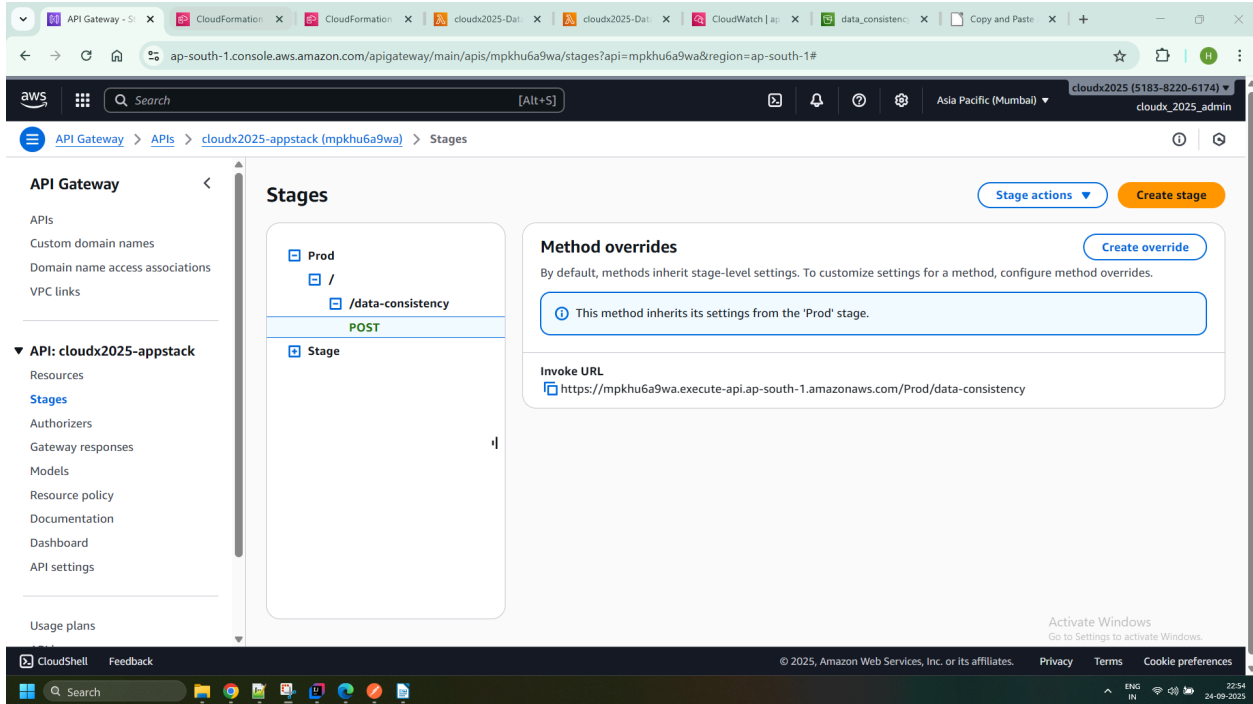
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Search

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1. From API Gateway



API Gateway - Stages x CloudFormation - Stages x CloudFormation | ap-south-1 x cloudx2025-DataConsistencyFunction x cloudx2025-DataConsistencyFunction x CloudWatch | ap-south-1 x data_consistency.zip x +

ap-south-1.console.aws.amazon.com/cloudwatch/home?region=ap-south-1#logsV2:log-groups/log-group/\$252faws\$252fcloudx2025-DataConsistencyFunction...

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CloudWatch > Log groups > /aws/lambda/cloudx2025-DataConsistencyFunction > 2025/09/24/[\$LATEST]677addae00f44c42906d2541783ace7a

Log events

You can use the filter bar below to search for and match terms, phrases, or values in your log events. [Learn more about filter patterns](#)

Filter events - press enter to search Clear 1m 30m 1h 12h Custom Local timezone Display

Message

No older events at this moment. [Retry](#)

```
INIT_START Runtime Version: python:3.11.v93 Runtime Version ARN: arn:aws:lambda:ap-south-1::runtime:0517e723fa8dfc41b11b2324c8c497c137d4a3442c2e280cd9deb3ccfd45d8bd

START RequestId: c7228c9d-eaeb-47c1-8bf1-65227fab67e5 Version: $LATEST

Incoming event: {
  "resource": "/data-consistency",
  "path": "/data-consistency",
  "httpMethod": "POST",
  "headers": {
    "Accept": "*/*",
    "Accept-Encoding": "gzip, deflate, br",
    "CloudFront-Forwarded-Proto": "https",
    "CloudFront-Is-Desktop-Viewer": "true",
    "CloudFront-Is-Mobile-Viewer": "false"
```

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Search 22:44 24-09-2025

ap-south-1.console.aws.amazon.com/cloudwatch/home?region=ap-south-1#logsV2:log-groups/log-group/\$252faws\$252fcloudx2025-DataConsistencyFunction...

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CloudWatch > Log groups > /aws/lambda/cloudx2025-DataConsistencyFunction > 2025/09/24/[\$LATEST]677addae00f44c42906d2541783ace7a

Log events

You can use the filter bar below to search for and match terms, phrases, or values in your log events. [Learn more about filter patterns](#)

Filter events - press enter to search Clear 1m 30m 1h 12h Custom Local timezone Display

Message

```
,
  "CloudFront-Viewer-Country": [
    "IN"
  ],
  "Host": [
    "mpkhu6a9wa.execute-api.ap-south-1.amazonaws.com"
  ],
  "Postman-Token": [
    "969248ea-71dd-46e6-9909-14d9b9ce1888"
  ],
  "User-Agent": [
    "PostmanRuntime/7.47.1"
  ],
}
```

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Search 22:44 24-09-2025

ap-south-1.console.aws.amazon.com/cloudwatch/home?region=ap-south-1#logsV2:log-groups/log-group/\$252Faws\$252Flambda\$252Fcloudx2025-DataConsistencyFunction...

Log events

You can use the filter bar below to search for and match terms, phrases, or values in your log events. [Learn more about filter patterns](#)

Filter events - press enter to search

Clear 1m 30m 1h 12h Custom Local timezone Display

Message

```
"userAgent": "PostmanRuntime/7.47.1",
"user": null
},
"domainName": "mpkhu6a9wa.execute-api.ap-south-1.amazonaws.com",
"deploymentId": "1gg8ao",
"apiId": "mpkhu6a9wa"
},
"body": null,
"isBase64Encoded": false
}

Data consistency check result: {'consistent': True, 'db_images_count': 0, 's3_images_count': 0, 'missing_in_s3': [], 'extra_in_s3': []}

END RequestId: c7228c9d-eaeb-47c1-8bf1-65227fab67e5

REPORT RequestId: c7228c9d-eaeb-47c1-8bf1-65227fab67e5 Duration: 356.58 ms Billed Duration: 1991 ms Memory Size: 128 MB Max Memory Used: 98 MB Init Duration: 1634.18 ms

START RequestId: c7228c9d-eaeb-47c1-8bf1-65227fab67e5 Version: 6.1 ATECT
```

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22:45 24-09-2025

2. From WebApp

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Invite Upgrade

cloudx2025-app /notification/data-consistency

POST {{cloudx-app-url}}/notification/data-consistency

Send

Params Authorization Headers (7) Body Scripts Settings Cookies

Query Params

Key	Value	Description	Bulk Edit
Key	Value	Description	

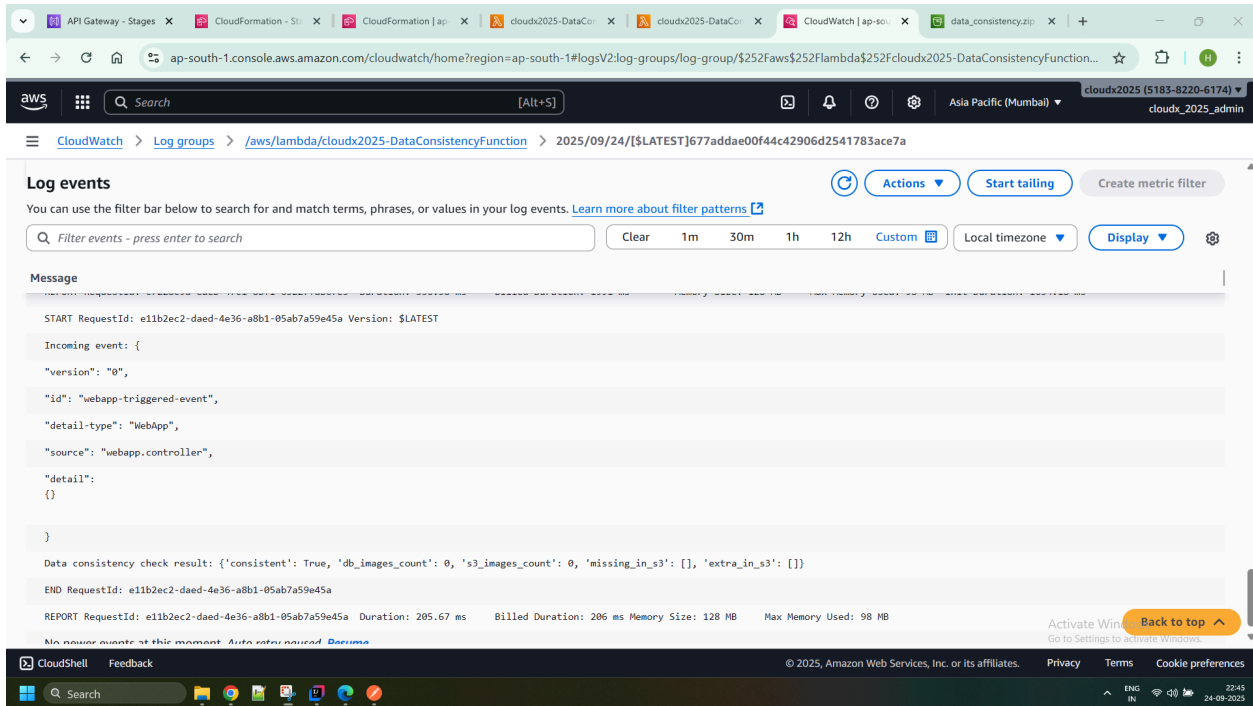
Body Cookies Headers (4) Test Results

200 OK 321 ms 286 B Save Response

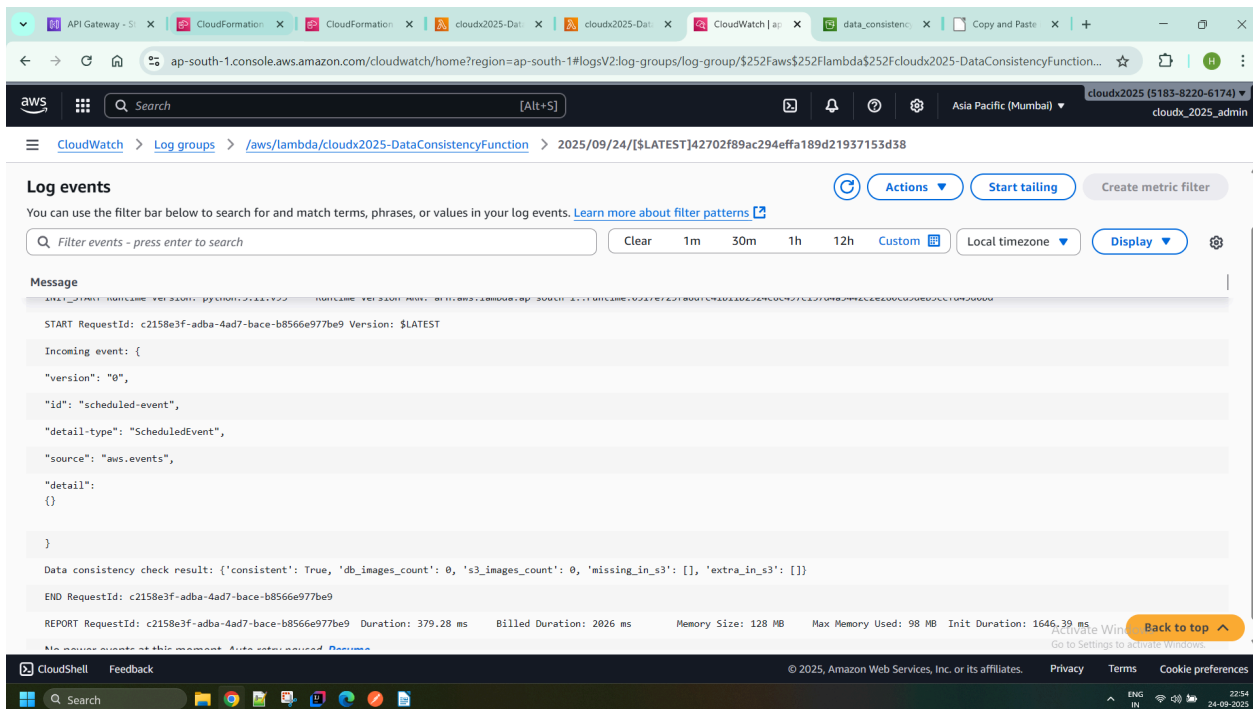
JSON Preview Visualize

```
1 {
2   "statusCode": 200,
3   "body": "{\"consistent\": true, \"db_images_count\": 0, \"s3_images_count\": 0, \"missing_in_s3\": [], \"extra_in_s3\": []}"
4 }
```

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3. From EventBridge (scheduled @ 20 mins interval)



CloudFormation scripts

1. Network stack

Parameters:

```
ProjectName:
  Type: String
BucketSuffix:
  Type: String
  Description: "Suffix for the S3 bucket name"
DBUsername:
  Type: String
  NoEcho: true
  Description: "Master username for RDS"
DBPassword:
  Type: String
  NoEcho: true
  Description: "Master password for RDS"
DBName:
  Type: String
  Default: cloudx2025
  Description: "Name of the database to create"

Resources:
  S3Bucket:
    Type: AWS::S3::Bucket
    Properties:
      BucketName: !Sub "cloudx2025-${BucketSuffix}"

  VPC:
    Type: AWS::EC2::VPC
    Properties:
      CidrBlock: 10.0.0.0/16
      EnableDnsHostnames: true
      EnableDnsSupport: true
      Tags:
        - Key: Name
          Value: !Sub "${ProjectName}-VPC"

  InternetGateway:
    Type: AWS::EC2::InternetGateway
    Properties:
      Tags:
        - Key: Name
          Value: !Sub "${ProjectName}-IGW"

  AttachGateway:
    Type: AWS::EC2::VPCGatewayAttachment
    Properties:
      VpcId: !Ref VPC
      InternetGatewayId: !Ref InternetGateway

  PublicSubnetA:
    Type: AWS::EC2::Subnet
```

```
Properties:
  VpcId: !Ref VPC
  CidrBlock: 10.0.1.0/24
  AvailabilityZone: !Select [0, !GetAZs '']
  MapPublicIpOnLaunch: true
  Tags:
    - Key: Name
      Value: !Sub "${ProjectName}-PublicSubnet-A"
```

```
PublicSubnetB:
  Type: AWS::EC2::Subnet
  Properties:
    VpcId: !Ref VPC
    CidrBlock: 10.0.2.0/24
    AvailabilityZone: !Select [1, !GetAZs '']
    MapPublicIpOnLaunch: true
    Tags:
      - Key: Name
        Value: !Sub "${ProjectName}-PublicSubnet-B"
```

```
PrivateSubnetA:
  Type: AWS::EC2::Subnet
  Properties:
    VpcId: !Ref VPC
    CidrBlock: 10.0.3.0/24
    AvailabilityZone: !Select [0, !GetAZs ''] # This will be ap-south-1a
    MapPublicIpOnLaunch: false
    Tags:
      - Key: Name
        Value: !Sub "${ProjectName}-PrivateSubnetA"
```

```
PrivateSubnetB:
  Type: AWS::EC2::Subnet
  Properties:
    VpcId: !Ref VPC
    CidrBlock: 10.0.4.0/24
    AvailabilityZone: !Select [1, !GetAZs ''] # Changed to index 1 for
ap-south-1b
    MapPublicIpOnLaunch: false
    Tags:
      - Key: Name
        Value: !Sub "${ProjectName}-PrivateSubnetB"
```

```
PublicRouteTable:
  Type: AWS::EC2::RouteTable
  Properties:
    VpcId: !Ref VPC
    Tags:
      - Key: Name
```

```

        Value: !Sub "${ProjectName}-PublicRT"

PublicRoute:
  Type: AWS::EC2::Route
  DependsOn: AttachGateway
  Properties:
    RouteTableId: !Ref PublicRouteTable
    DestinationCidrBlock: 0.0.0.0/0
    GatewayId: !Ref InternetGateway

PublicSubnetARouteTableAssociation:
  Type: AWS::EC2::SubnetRouteTableAssociation
  Properties:
    SubnetId: !Ref PublicSubnetA
    RouteTableId: !Ref PublicRouteTable

PublicSubnetBRouteTableAssociation:
  Type: AWS::EC2::SubnetRouteTableAssociation
  Properties:
    SubnetId: !Ref PublicSubnetB
    RouteTableId: !Ref PublicRouteTable

PrivateRouteTable:
  Type: AWS::EC2::RouteTable
  Properties:
    VpcId: !Ref VPC
    Tags:
      - Key: Name
        Value: !Sub "${ProjectName}-PrivateRT"

PrivateSubnetARouteTableAssociation:
  Type: AWS::EC2::SubnetRouteTableAssociation
  Properties:
    SubnetId: !Ref PrivateSubnetA
    RouteTableId: !Ref PrivateRouteTable

PrivateSubnetBRouteTableAssociation:
  Type: AWS::EC2::SubnetRouteTableAssociation
  Properties:
    SubnetId: !Ref PrivateSubnetB
    RouteTableId: !Ref PrivateRouteTable

S3VpcEndpoint:
  Type: AWS::EC2::VPCEndpoint
  Properties:
    VpcEndpointType: Gateway
    ServiceName: !Sub "com.amazonaws.${AWS::Region}.s3"
    VpcId: !Ref VPC
    RouteTableIds:

```

- !Ref PrivateRouteTable

DBSubnetGroup:

Type: AWS::RDS::DBSubnetGroup

Properties:

DBSubnetGroupDescription: !Sub "\${ProjectName} RDS subnet group"

SubnetIds:

- !Ref PrivateSubnetA
- !Ref PrivateSubnetB

DBSubnetGroupName: !Sub "\${ProjectName}-db-subnet"

AppSecurityGroup:

Type: AWS::EC2::SecurityGroup

Properties:

GroupDescription: "Allow app traffic from ALB and SSH access"

VpcId: !Ref VPC

SecurityGroupIngress:

- IpProtocol: tcp
FromPort: 22
ToPort: 22
CidrIp: 103.41.39.253/32

Tags:

- Key: Name
Value: !Sub "\${ProjectName}-AppSecurityGroup"

RdsSecurityGroup:

Type: AWS::EC2::SecurityGroup

Properties:

GroupDescription: "Allow MySQL from app"

VpcId: !Ref VPC

SecurityGroupIngress:

- IpProtocol: tcp
FromPort: 3306
ToPort: 3306
SourceSecurityGroupId: !Ref AppSecurityGroup

MyDB:

Type: AWS::RDS::DBInstance

Properties:

Engine: mysql

DBInstanceClass: db.t3.micro

AllocatedStorage: 20

DBName: !Ref DBName

MasterUsername: !Ref DBUsername

MasterUserPassword: !Ref DBPassword

PubliclyAccessible: false

MultiAZ: true

BackupRetentionPeriod: 0 # Disable automated backups

VPCSecurityGroups:

```
    - !Ref RdsSecurityGroup
    DBSubnetGroupName: !Ref DBSubnetGroup
    DeletionProtection: false
```

Outputs:

```
VpcId:
  Value: !Ref VPC
  Export:
    Name: !Sub "${ProjectName}-VpcId"
PublicSubnetAId:
  Value: !Ref PublicSubnetA
  Export:
    Name: !Sub "${ProjectName}-PublicSubnetAId"
PublicSubnetBId:
  Value: !Ref PublicSubnetB
  Export:
    Name: !Sub "${ProjectName}-PublicSubnetBId"
PrivateSubnetAId:
  Value: !Ref PrivateSubnetA
  Export:
    Name: !Sub "${ProjectName}-PrivateSubnetAId"
PrivateSubnetBId:
  Value: !Ref PrivateSubnetB
  Export:
    Name: !Sub "${ProjectName}-PrivateSubnetBId"
S3BucketName:
  Value: !Ref S3Bucket
  Export:
    Name: !Sub "${ProjectName}-S3BucketName"
RDSEndpoint:
  Value: !GetAtt MyDB.Endpoint.Address
  Export:
    Name: !Sub "${ProjectName}-RDSEndpoint"
RDSInstanceId:
  Value: !Ref MyDB
  Export:
    Name: !Sub "${ProjectName}-RDSInstanceId"
RDSUsername:
  Value: !Ref DBUsername
  Export:
    Name: !Sub "${ProjectName}-RDSUsername"
AppSecurityGroupId:
  Value: !Ref AppSecurityGroup
  Export:
    Name: !Sub "${ProjectName}-AppSecurityGroupId"
RDSDBName:
  Value: !Ref DBName
  Export:
    Name: !Sub "${ProjectName}-RDSDBName"
```

```
RdsSecurityGroupId:
  Value: !Ref RdsSecurityGroup
  Export:
    Name: !Sub "${ProjectName}-RdsSecurityGroupId"
```

2. App stack

```
AWSTemplateFormatVersion: '2010-09-09'
Transform: AWS::Serverless-2016-10-31
Description: Application stack with Lambda functions
```

```
Parameters:
  ProjectName:
    Type: String
  DeploymentBucket:
    Type: String
    Description: "S3 bucket name containing deployment artifacts (JAR, Lambda
code, layers)"
  AppJarKey:
    Type: String
    Description: "S3 key for the application JAR file"
  UploadsNotificationCodeKey:
    Type: String
    Default: "uploads_notification.zip"
    Description: "S3 key for the Uploads Notification Lambda function code"
  DataConsistencyCodeKey:
    Type: String
    Default: "data_consistency.zip"
    Description: "S3 key for the Data Consistency Lambda function code"
  LayerKey:
    Type: String
    Description: "S3 key for the Lambda layer zip file"
  RDSHost:
    Type: String
    Description: "RDS endpoint from network stack"
  DBUsername:
    Type: String
    Description: "Database username"
    NoEcho: true
  DBPassword:
    Type: String
    Description: "Database password"
    NoEcho: true
  S3BucketName:
    Type: String
    Description: "S3 bucket name from network stack"
  DBName:
    Type: String
    Description: "Database name from network stack"
```

Resources:

AppSecurityGroupIngress:

Type: AWS::EC2::SecurityGroupIngress

Properties:

GroupId:

Fn::ImportValue: !Sub "\${ProjectName}-AppSecurityGroupId"

IpProtocol: tcp

FromPort: 8080

ToPort: 8080

SourceSecurityGroupId: !Ref AlbSecurityGroup

AlbSecurityGroup:

Type: AWS::EC2::SecurityGroup

Properties:

GroupDescription: "Security group for ALB"

VpcId:

Fn::ImportValue: !Sub "\${ProjectName}-VpcId"

SecurityGroupIngress:

- IpProtocol: tcp

FromPort: 80

ToPort: 80

CidrIp: 0.0.0.0/0

Tags:

- Key: Name

Value: !Sub "\${ProjectName}-ALBSecurityGroup"

AppLaunchTemplate:

Type: AWS::EC2::LaunchTemplate

Properties:

LaunchTemplateData:

ImageId: ami-0b982602dbb32c5bd

InstanceType: t3.micro

KeyName: cloudx_ec2_keypair

IamInstanceProfile:

Name: cloudx2025_FullAccessRoleS3

SecurityGroupIds:

- Fn::ImportValue: !Sub "\${ProjectName}-AppSecurityGroupId"

UserData:

Fn::Base64: !Sub |

#!/bin/bash

yum install -y aws-cli

Install Java 17

amazon-linux-extras enable java-17-amazon-corretto

yum clean metadata

yum install -y java-17-amazon-corretto

Download and start the application

```

aws s3 cp s3://${DeploymentBucket}/${AppJarKey}
/home/ec2-user/cloudx2025-app.jar

# Create application service file with properly escaped environment
variables
cat << EOF > /etc/systemd/system/cloudx-app.service
[Unit]
Description=CloudX Application
After=network.target

[Service]
User=ec2-user
WorkingDirectory=/home/ec2-user
Environment=DB_HOST=${RDSTHost}
Environment=DB_USER=${DBUsername}
Environment=DB_PASS=${DBPassword}
Environment=DB_NAME=${DBName}
Environment=S3_BUCKET=${S3BucketName}
Environment=SERVER_PORT=8080
Environment=PROJECT_NAME=${ProjectName}

Environment=UPLOADS_NOTIFICATION_QUEUE_URL=${UploadsNotificationQueue}

Environment=UPLOADS_NOTIFICATION_TOPIC_ARN=${UploadsNotificationTopic}
ExecStart=/usr/bin/java -jar cloudx2025-app.jar
SuccessExitStatus=143
TimeoutStopSec=10
Restart=on-failure
RestartSec=5

[Install]
WantedBy=multi-user.target
EOF

# Log the environment variables for debugging
echo "Environment variables being used:" > /home/ec2-user/env.log
echo "DB_HOST=${RDSTHost}" >> /home/ec2-user/env.log
echo "DB_USER=${DBUsername}" >> /home/ec2-user/env.log
echo "DB_NAME=${DBName}" >> /home/ec2-user/env.log
echo "S3_BUCKET=${S3BucketName}" >> /home/ec2-user/env.log
echo "SERVER_PORT=8080" >> /home/ec2-user/env.log

# Set permissions and start service
chmod 644 /etc/systemd/system/cloudx-app.service
systemctl daemon-reload
systemctl enable cloudx-app
systemctl start cloudx-app

# Save logs to persistent location

```

```
ln -sf /var/log/messages /home/ec2-user/system.log
touch /home/ec2-user/app.log
chown ec2-user:ec2-user /home/ec2-user/app.log
journalctl -u cloudx-app -f > /home/ec2-user/app.log &
```

AppAutoScalingGroup:

Type: AWS::AutoScaling::AutoScalingGroup

Properties:

VPCZoneIdentifier:

- Fn::ImportValue: !Sub "\${ProjectName}-PublicSubnetAId"
- Fn::ImportValue: !Sub "\${ProjectName}-PublicSubnetBId"

LaunchTemplate:

LaunchTemplateId: !Ref AppLaunchTemplate

Version: !GetAtt AppLaunchTemplate.LatestVersionNumber

MinSize: "1"

MaxSize: "3"

DesiredCapacity: "1"

TargetGroupARNs:

- !Ref AppTargetGroup

Tags:

- Key: Name

Value: !Sub "\${ProjectName}-AppInstance"

PropagateAtLaunch: true

AppTargetGroup:

Type: AWS::ElasticLoadBalancingV2::TargetGroup

Properties:

VpcId:

Fn::ImportValue: !Sub "\${ProjectName}-VpcId"

Port: 8080

Protocol: HTTP

TargetType: instance

HealthCheckProtocol: HTTP

HealthCheckPort: "8080"

HealthCheckPath: "/"

HealthCheckIntervalSeconds: 15

HealthCheckTimeoutSeconds: 5

HealthyThresholdCount: 2

UnhealthyThresholdCount: 5

Matcher:

HttpCode: 200

Name: !Sub "\${ProjectName}-tg"

AppLoadBalancer:

Type: AWS::ElasticLoadBalancingV2::LoadBalancer

Properties:

Subnets:

- Fn::ImportValue: !Sub "\${ProjectName}-PublicSubnetAId"
- Fn::ImportValue: !Sub "\${ProjectName}-PublicSubnetBId"

```

SecurityGroups:
  - !Ref AlbSecurityGroup
Tags:
  - Key: Name
    Value: !Sub "${ProjectName}-LoadBalancer"

AppListener:
  Type: AWS::ElasticLoadBalancingV2::Listener
  Properties:
    DefaultActions:
      - Type: forward
        TargetGroupArn: !Ref AppTargetGroup
    LoadBalancerArn: !Ref AppLoadBalancer
    Port: 80
    Protocol: HTTP

UploadsNotificationQueue:
  Type: AWS::SQS::Queue
  Properties:
    QueueName: !Sub "${ProjectName}-UploadsNotificationQueue"

UploadsNotificationTopic:
  Type: AWS::SNS::Topic
  Properties:
    TopicName: !Sub "${ProjectName}-UploadsNotificationTopic"

UploadsNotificationLambdaRole:
  Type: AWS::IAM::Role
  Properties:
    RoleName: !Sub "${ProjectName}-UploadsNotificationLambdaRole"
    AssumeRolePolicyDocument:
      Version: "2012-10-17"
      Statement:
        - Effect: Allow
          Principal:
            Service: lambda.amazonaws.com
          Action: sts:AssumeRole
    ManagedPolicyArns:
      - arn:aws:iam::aws:policy/service-role/AWSLambdaBasicExecutionRole
    Policies:
      - PolicyName: SNSPublishPolicy
        PolicyDocument:
          Version: "2012-10-17"
          Statement:
            - Effect: Allow
              Action: sns:Publish
              Resource: !Ref UploadsNotificationTopic
      - PolicyName: SQSReceivePolicy
        PolicyDocument:

```

```
Version: "2012-10-17"
Statement:
  - Effect: Allow
    Action:
      - sqs:ReceiveMessage
      - sqs>DeleteMessage
      - sqs:GetQueueAttributes
    Resource: !GetAtt UploadsNotificationQueue.Arn
```

UploadsNotificationFunction:

Type: AWS::Serverless::Function

Properties:

```
FunctionName: !Sub "${ProjectName}-UploadsNotificationFunction"
Runtime: python3.11
Handler: uploads_notification.lambda_handler
Role: !GetAtt UploadsNotificationLambdaRole.Arn
CodeUri:
  Bucket: !Ref DeploymentBucket
  Key: !Ref UploadsNotificationCodeKey
Environment:
  Variables:
    SNS_TOPIC_ARN: !Ref UploadsNotificationTopic
Timeout: 30
```

SQS trigger for Lambda

UploadsNotificationLambdaTrigger:

Type: AWS::Lambda::EventSourceMapping

Properties:

```
FunctionName: !Ref UploadsNotificationFunction
EventSourceArn: !GetAtt UploadsNotificationQueue.Arn
BatchSize: 10
Enabled: true
```

DataConsistencyLambdaRole:

Type: AWS::IAM::Role

Properties:

```
RoleName: !Sub "${ProjectName}-DataConsistencyLambdaRole"
AssumeRolePolicyDocument:
  Version: "2012-10-17"
  Statement:
    - Effect: Allow
      Principal:
        Service: lambda.amazonaws.com
      Action: sts:AssumeRole
```

ManagedPolicyArns:

- arn:aws:iam::aws:policy/service-role/AWSLambdaBasicExecutionRole
- arn:aws:iam::aws:policy/service-role/AWSLambdaVPCAccessExecutionRole
- arn:aws:iam::aws:policy/AmazonS3FullAccess
- arn:aws:iam::aws:policy/AmazonRDSFullAccess

```

DataConsistencySecurityGroup:
  Type: AWS::EC2::SecurityGroup
  Properties:
    GroupDescription: "Security group for Data Consistency Lambda"
    VpcId:
      Fn::ImportValue: !Sub "${ProjectName}-VpcId"

RdsSecurityGroupIngressFromLambda:
  Type: AWS::EC2::SecurityGroupIngress
  Properties:
    GroupId:
      Fn::ImportValue: !Sub "${ProjectName}-RdsSecurityGroupId"
    IpProtocol: tcp
    FromPort: 3306
    ToPort: 3306
    SourceSecurityGroupId: !Ref DataConsistencySecurityGroup

LambdaInvokeInlinePolicy:
  Type: AWS::IAM::Policy
  Properties:
    PolicyName: !Sub "${ProjectName}-LambdaInvokeInlinePolicy"
    PolicyDocument:
      Version: "2012-10-17"
      Statement:
        - Effect: Allow
          Action: lambda:InvokeFunction
          Resource: !GetAtt DataConsistencyFunction.Arn
    Roles:
      - cloudx2025_FullAccessRoleS3

DataConsistencyLayer:
  Type: AWS::Lambda::LayerVersion
  Properties:
    LayerName: !Sub "${ProjectName}-DataConsistencyLayer"
    CompatibleRuntimes:
      - python3.11
    Content:
      S3Bucket: !Ref DeploymentBucket
      S3Key: !Ref LayerKey

DataConsistencyFunction:
  Type: AWS::Serverless::Function
  Properties:
    FunctionName: !Sub "${ProjectName}-DataConsistencyFunction"
    Runtime: python3.11
    Handler: data_consistency.lambda_handler
    Role: !GetAtt DataConsistencyLambdaRole.Arn
    Layers:

```

```

    - !Ref DataConsistencyLayer
VpcConfig:
  SecurityGroupIds:
    - !Ref DataConsistencySecurityGroup
  SubnetIds:
    - Fn::ImportValue: !Sub "${ProjectName}-PrivateSubnetAId"
    - Fn::ImportValue: !Sub "${ProjectName}-PrivateSubnetBId"
Environment:
  Variables:
    DB_HOST: !Ref RDSTHost
    DB_USER: !Ref DBUsername
    DB_PASS: !Ref DBPassword
    DB_NAME: !Ref DBName
    S3_BUCKET: !Ref S3BucketName
CodeUri:
  Bucket: !Ref DeploymentBucket
  Key: !Ref DataConsistencyCodeKey
Events:
  DataConsistencySchedule:
    Type: Schedule
    Properties:
      Schedule: rate(20 minutes)
      Name: !Sub "${ProjectName}-DataConsistencySchedule"
      Description: "Schedule to run data consistency check every 20
minutes"
      Input: |
        {
          "version": "0",
          "id": "scheduled-event",
          "detail-type": "ScheduledEvent",
          "source": "aws.events",
          "detail": {}
        }
  DataConsistencyApi:
    Type: Api
    Properties:
      Path: /data-consistency
      Method: post

Outputs:
  AppAutoScalingGroupName:
    Value: !Ref AppAutoScalingGroup
    Export:
      Name: !Sub "${ProjectName}-AppAutoScalingGroupName"
  AppLoadBalancerDNS:
    Value: !GetAtt AppLoadBalancer.DNSName
    Export:
      Name: !Sub "${ProjectName}-AppLoadBalancerDNS"
  UploadsNotificationQueueArn:

```

```
    Description: "ARN of the uploads notification SQS queue"
    Value: !GetAtt UploadsNotificationQueue.Arn
    Export:
      Name: !Sub "${ProjectName}-UploadsNotificationQueueArn"
UploadsNotificationTopicArn:
    Description: "ARN of the uploads notification SNS topic"
    Value: !Ref UploadsNotificationTopic
    Export:
      Name: !Sub "${ProjectName}-UploadsNotificationTopicArn"
UploadsNotificationQueueUrl:
    Description: "URL of the uploads notification SQS queue"
    Value: !Ref UploadsNotificationQueue
    Export:
      Name: !Sub "${ProjectName}-UploadsNotificationQueueUrl"
```